

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	17 Apr 2015	First Release	Muhammed Shameem
01	2 Oct 2016	Template update & Content update	ONG KEAN SHENG

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

1. Remove 2 x screws from top enclosure of the X-Ray machine as shown in the Figure 1.

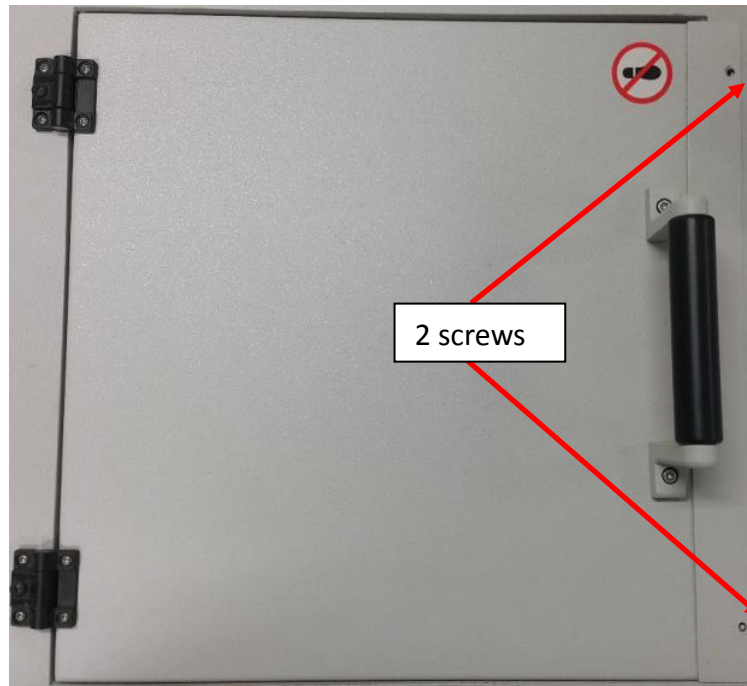


Figure 1 X-ray top enclosure

2. Remove SHCS M6×60 (4x) from the top of the x-ray tube as shown.

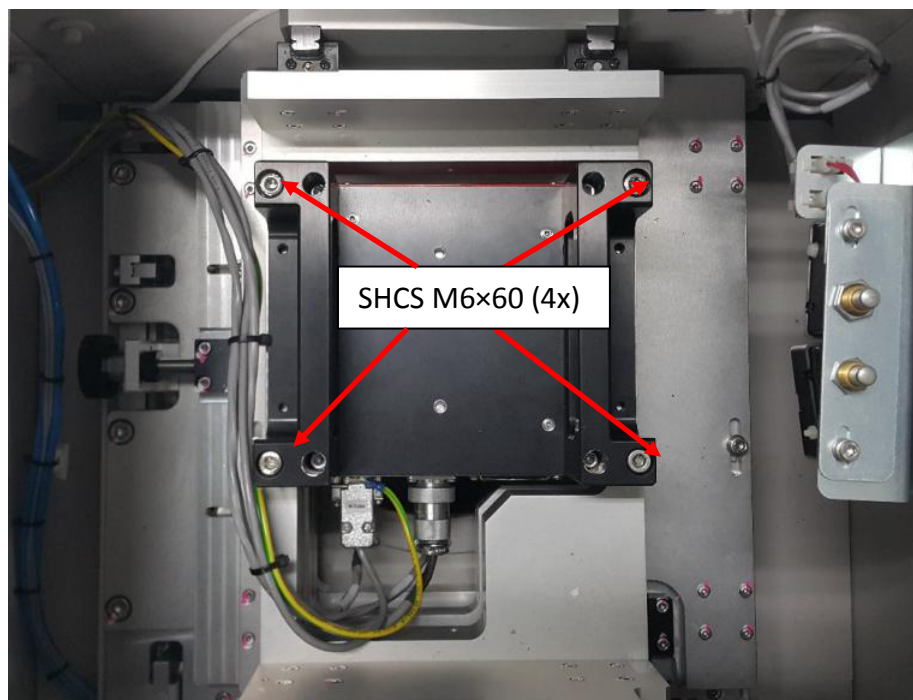


Figure 2 X-ray tube holder

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

- Lift the tube and place it at the top of the machine as shown in Figure 3.

Remarks: Caution on cable assembly upon positioning or removal

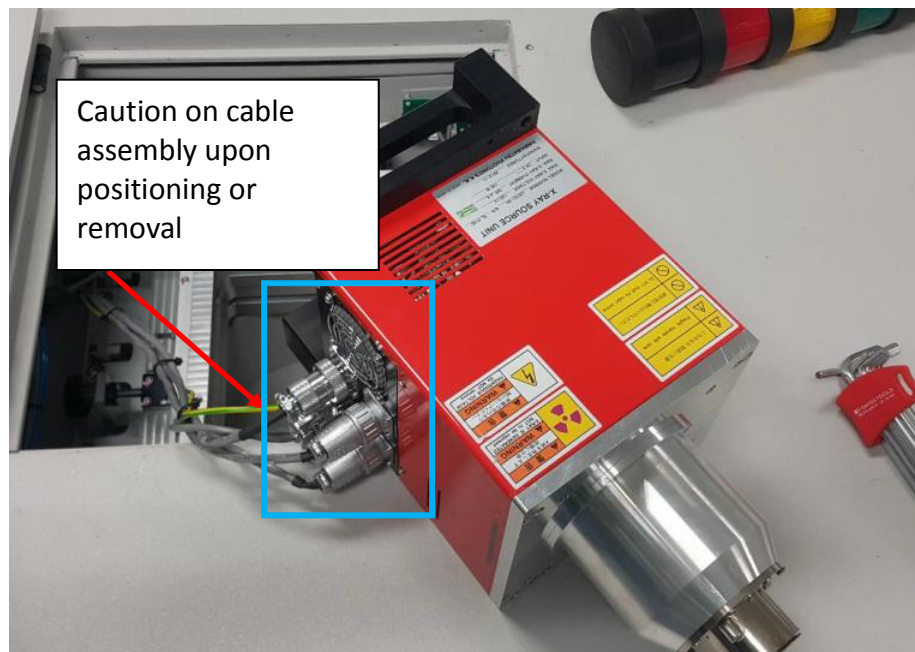


Figure 3 X-ray top positioning

- Remove 4 x Screws(M5×20) from the x-ray tube holder as shown in the Figure 4.



Figure 4 X-ray tube holder

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

5. After remove SHCS M5×20 (4x), remove X-ray tube holder as shown in the Figure 5.

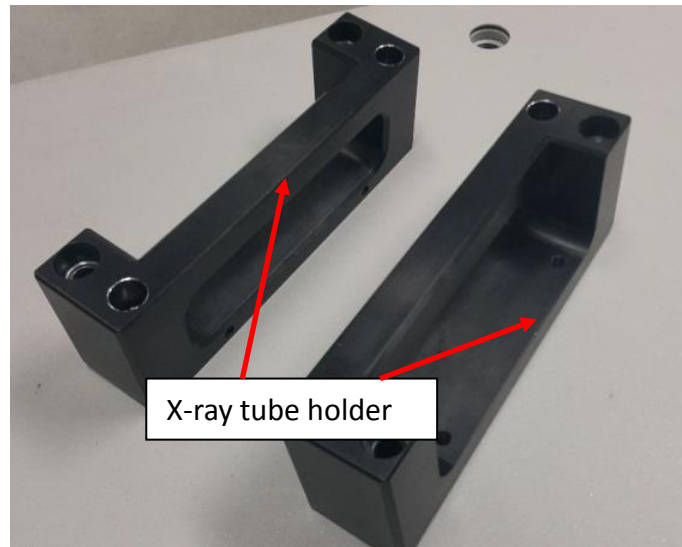


Figure 5 X-ray tube holder (2x)

6. Remove the following cables from the X-ray tube.

- i. Interlock 1
- ii. Interlock 2
- iii. 24VDC
- iv. RS-232C / H-tube cable
- v. GND

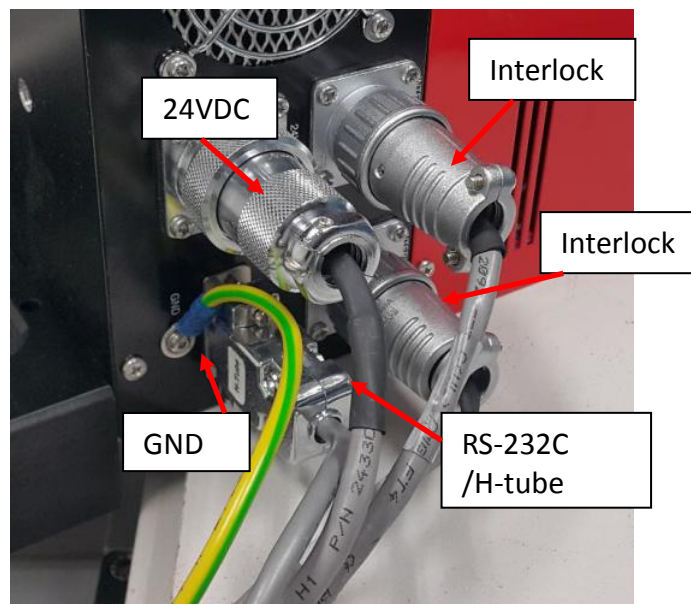


Figure 6 X-ray tube cables

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

7. Close the tube target with cover plate as shown in Figure 7.

Remarks : May obtain from new tube packaging.



Figure 7 Tube target cover plate

8. Gently pack the defective X-ray tube inside box.

9. Visual check X-ray chamber to detect any drop off parts.

E.g. Screws, Nut and etc



Figure 8 X-Ray Chamber

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

10. Prepare New X ray Tube.

11. Mount the X-ray tube holder using the SHCS M5×20 (4x) as shown in the Figure 8.

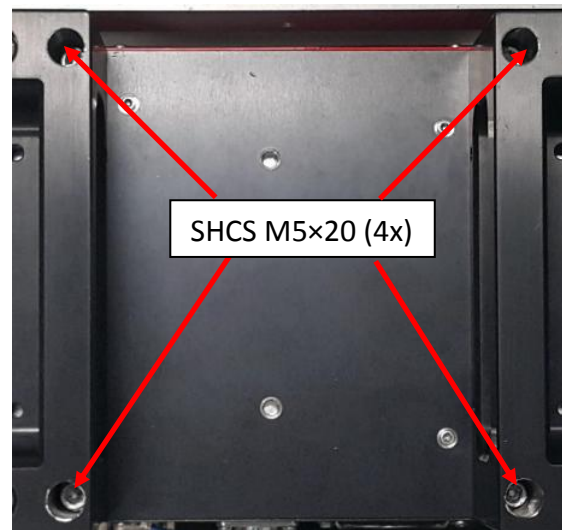


Figure 9 X-ray tube holder

12. Insert the following cables to the X-ray tube as shown in Figure 10 and Figure 11.

- vi. Interlock 1
- vii. Interlock 2
- viii. 24VDC
- ix. RS-232C / H-tube cable
- x. GND

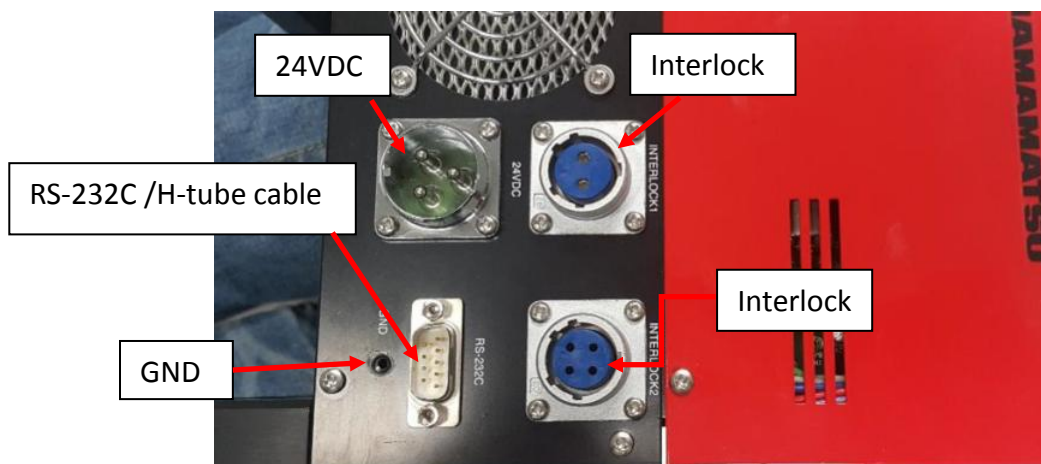


Figure 10 X-ray tube cables (1)

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

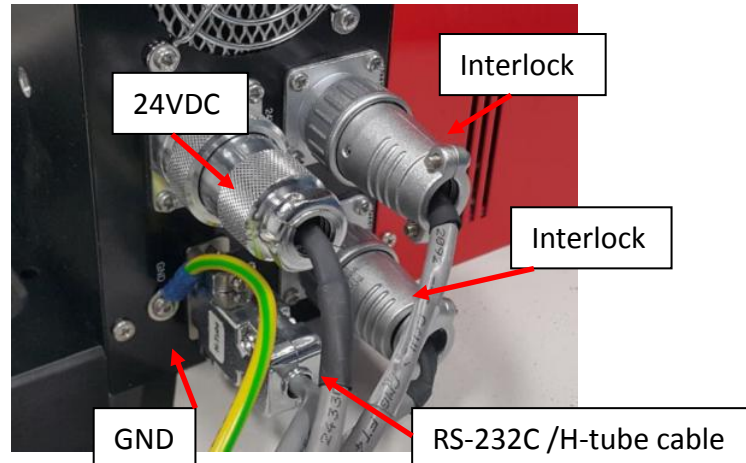


Figure 11 X-ray tube cables (2)

13. Open cover plate from new tube target as shown in Figure 12.

May ignore this step if cover plate had been transfer at Step 7.



Figure 12 Tube target cover plate

14. Insert X-ray tube as shown in Figure 13 and tighten X-ray tube by using SHCS M6×60 (4x)

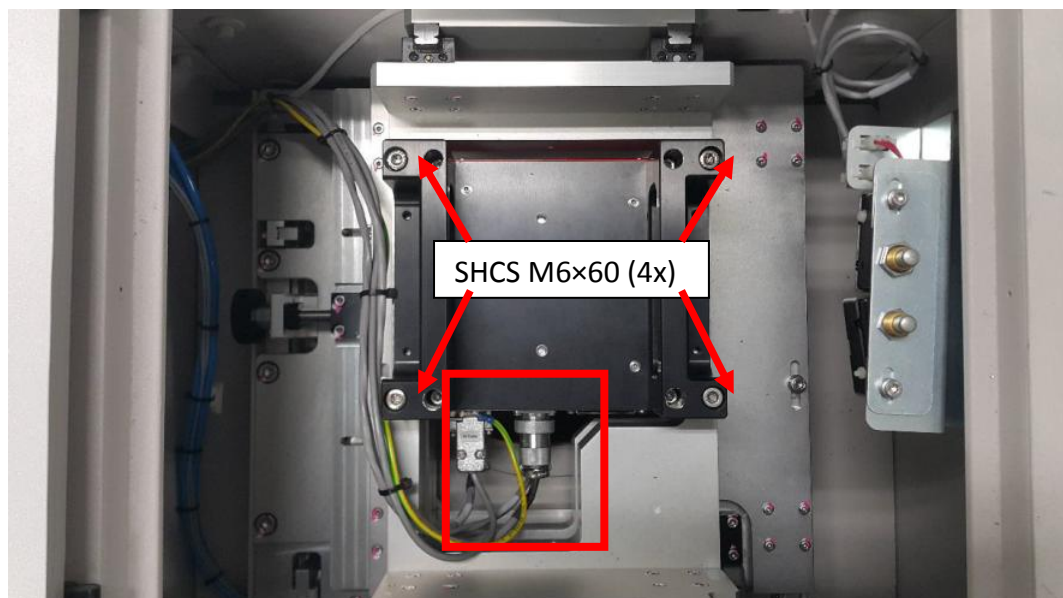


Figure 13 X-ray tube placement

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

15. Using cable tie to secure cables on the X-ray tube holder as shown in Figure 14.

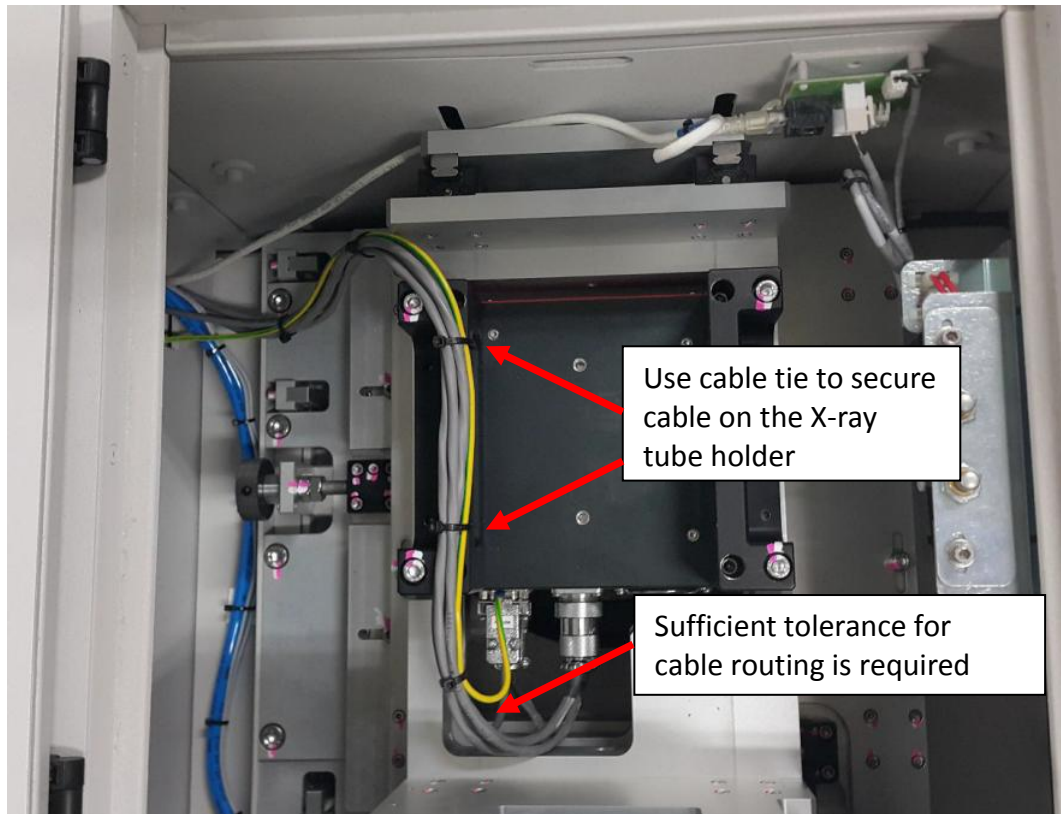


Figure 14 X-ray tube placement

16. Close top enclosure by apply 2 screws as shown in the Figure 15.

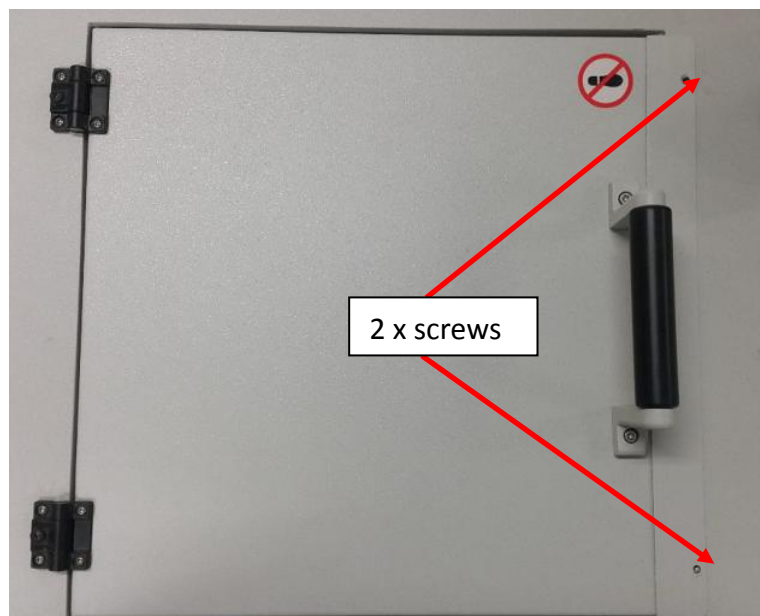


Figure 15 Top enclosure

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

17. Turn off air pressure regulator.

18. Manually exercise inner barrier shuttle plate and visual check either any friction between inner barrier shuttle plate & tube target.



Figure 16 Inner barrier shuttle plate & tube target

19. Close all access door.

20. Turn on air pressure regulator and set it to 80 PSI.

21. Go to X ray Source Tab and click “ Initialize”.

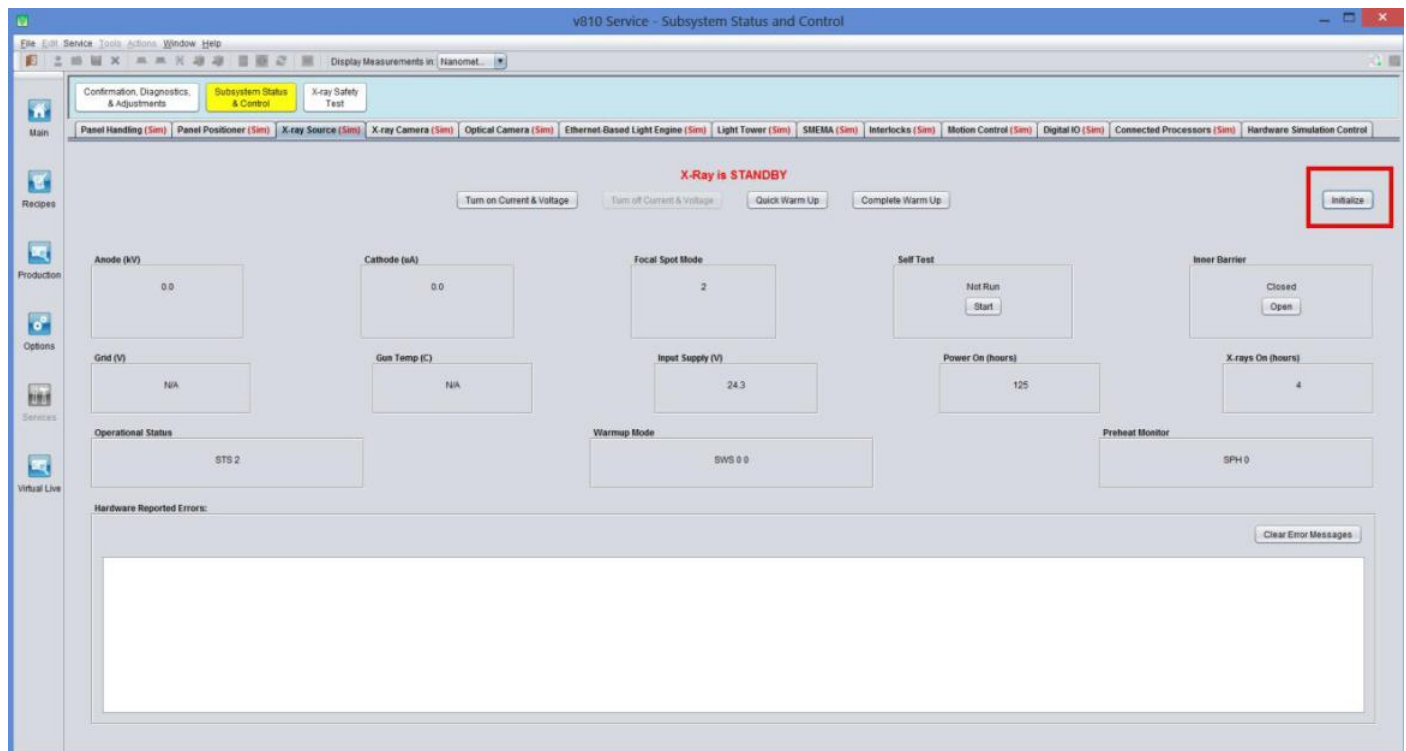


Figure 17 X ray Source

Remarks : Tube ramp will last for 90 Min (approximately)

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Ray Tube Replacement Procedure	Effective Date	2 Oct 2016

22. Run CD&A 10 times.

23. Go to the following directory **C:\Program Files\AXI System\v810\REV\log\calib\XraySpot** and obtain the cam_0.png. Ensure the image of the X-ray spot is optimized as shown in Figure 18.

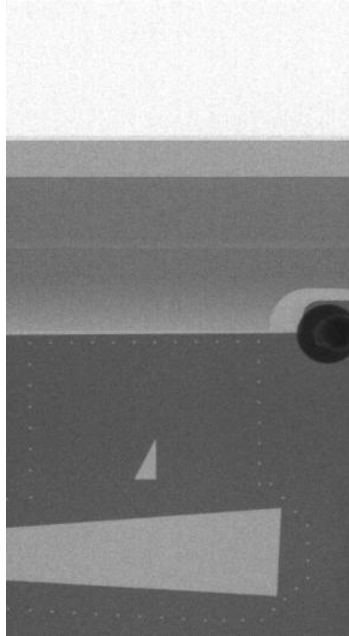


Figure 18 X-ray spot Golden Unit

24. Go to the following directory **C:\Program Files\AXI System\v810\REV\log\calib\Magnification** and obtain the latest magnification.log and highMagnification.log file.

i. magnification.log.

The magnification value should be within this range:

➤ Standard V810 machine : 5.20-5.25

ii. highMagnification.log (applicable for machine with VM option only)

The magnification value should be within this range:

➤ Standard V810 machine : 9.07-9.09

25. If the range is not within the mentioned specification in Step 24, may refer to **AXI STD V810 VM Tube Height Adjustment Procedure** for tube height adjustment.